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Note I.1 — Iitaka Fibration

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These notes give a first introduction to the Iitaka fibration. The guiding point is that a variety with non-negative Kodaira dimension carries a canonical rational map whose base has dimension $\kappa(X)$ and whose very general fibers have Kodaira dimension zero. As we will see Iitaka fibration is one of the building block of minimal model program. Iitaka fibration is also very useful when dealing with problem with intermediate Kodaira dimension. Iitaka fibration is also very useful in the “divide and rule” construction: it isolates the positive Kodaira dimension on the base and leaves a Kodaira-dimension-zero problem on the general fiber.

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1 Introduction

Let X be a normal projective variety, and suppose for the moment that $\kappa(X) \geq 0$. The pluricanonical systems $|mK_X|$ define rational maps

$$\Phi_{|mK_X|}: X \dashrightarrow \mathbb{P}H^0(X, mK_X)^\vee.$$

For m sufficiently large and divisible, these maps stabilize up to birational equivalence. The resulting birational fiber space is the *Iitaka fibration* of X :

$$X' \longrightarrow Y, \quad \dim Y = \kappa(X), \quad \kappa(F) = 0$$

for a very general fiber F .

Iitaka fibration as basic building block in the minimal model progoe. The minimal model program suggests the following coarse dichotomy. If $\kappa(X) < 0$, the expected output is a Mori fiber space. If $\kappa(X) \geq 0$, the expected output is a minimal model whose canonical divisor is semiample, and

hence induces the Iitaka fibration onto the canonical model Y :

$$\begin{array}{ccc}
 & W & \\
 \swarrow & & \searrow \\
 X & \dashrightarrow & X_{\min} \\
 & & \downarrow g \\
 & & Y.
 \end{array}$$

In the minimal model program, $K_{X_{\min}} \sim_{\mathbb{Q}} g^* A$ for an ample $\mathbb{Q}$ -divisor A on Y , and the general fiber has Kodaira dimension 0.

2 Kodaira Maps and Iitaka–Kodaira Dimension

2.1 Kodaira maps

Definition 2.1 (Kodaira map). Let X be a normal projective variety and let D be a **Cartier divisor** on X with $H^0(X, \mathcal{O}_X(D)) \neq 0$. The complete linear system $|D|$ defines a rational map

$$\Phi_{|D|}: X \dashrightarrow \mathbb{P}H^0(X, \mathcal{O}_X(D))^\vee.$$

If $s_0, \dots, s_N$ is a basis of $H^0(X, \mathcal{O}_X(D))$, then away from the base locus this map is

$$x \mapsto [s_0(x) : \dots : s_N(x)].$$

Equivalently, x is sent to the hyperplane of sections vanishing at x .

Remark 2.2. The map is undefined exactly at the base locus $\text{Bs } |D|$, where every section of $\mathcal{O}_X(D)$ vanishes (but it's possible to extend across the indeterminacy locus sometimes). After replacing X by a resolution of the graph of $\Phi_{|D|}$, the Kodaira map becomes a morphism. This replacement is harmless for birational questions such as Kodaira dimension and the Iitaka fibration.

Definition 2.3 (Iitaka dimension of a divisor). Let L be a line bundle on a normal projective variety X . Set

$$\mathbb{N}(X, L) = \{m \in \mathbb{N} \mid H^0(X, L^{\otimes m}) \neq 0\},$$

which forms a semi-group. If $\mathbb{N}(X, L) = \emptyset$, define $\kappa(X, L) = -\infty$. Otherwise,

$$\kappa(X, L) = \max_{m \in \mathbb{N}(X, L)} \dim \Phi_{|mL|}(X),$$

here image means closure of image of the domain of definition, we will omit the closure in the rest of the note. For a smooth projective variety X , the Kodaira dimension is $\kappa(X) = \kappa(X, K_X)$. For a singular variety, $\kappa(X)$ is defined on a smooth birational model. And can be proved it's independent of choice of resolution. From the above definition, it's immediate to see that $\kappa(X, L) \leq \dim X$, when achieve the upper bound, we call L is big.

Example 2.4. Let X be a K3 surface and L a nef and big line bundle on X . For $m \gg 0$, the morphism $f := \varphi_{|mL|}$ contracts exactly the irreducible curves $C \subset X$ with $L \cdot C = 0$.

Negative of the curves being contracted. Since L is nef and big, $L^2 > 0$, so by the Hodge index theorem the intersection form is negative definite on the hyperplane $L^\perp = \{D \in \text{NS}(X) \otimes \mathbb{R} : D \cdot L = 0\}$: for $D \in L^\perp$ we have $D^2 \leq 0$, with equality only if $D \equiv 0$ numerically. Each contracted curve C_i satisfies $L \cdot C_i = 0$, so $C_i \in L^\perp$; since f contracts them to points, the classes $[C_i]$ are linearly independent. Hence for $D = \sum_i a_i C_i \neq 0$ we get $D^2 < 0$, that is, the intersection matrix $(C_i \cdot C_j)$ is negative definite.

Rational of the curves. Let C be one such curve. Since $K_X \cong \mathcal{O}_X$, genus formula gives $2\bar{g}(C) - 2 = C^2$. As C is effective and nonzero it is not numerically trivial, so by the above $C^2 < 0$; since the K3 lattice is even and $p_a(C) \geq 0$, this forces $C^2 = -2$ and $p_a(C) = 0$. An irreducible curve of arithmetic genus 0 is smooth, so $C \cong \mathbb{P}^1$: each contracted curve is a smooth rational (-2) -curve, and the exceptional locus is a union of configurations of such curves.

ADE singularities. Any connected configuration of smooth rational curves with negative definite intersection matrix and all self-intersections -2 has dual graph a simply-laced Dynkin diagram of type A_n , D_n , E_6 , E_7 , or E_8 (Du Val); see Figure 1.

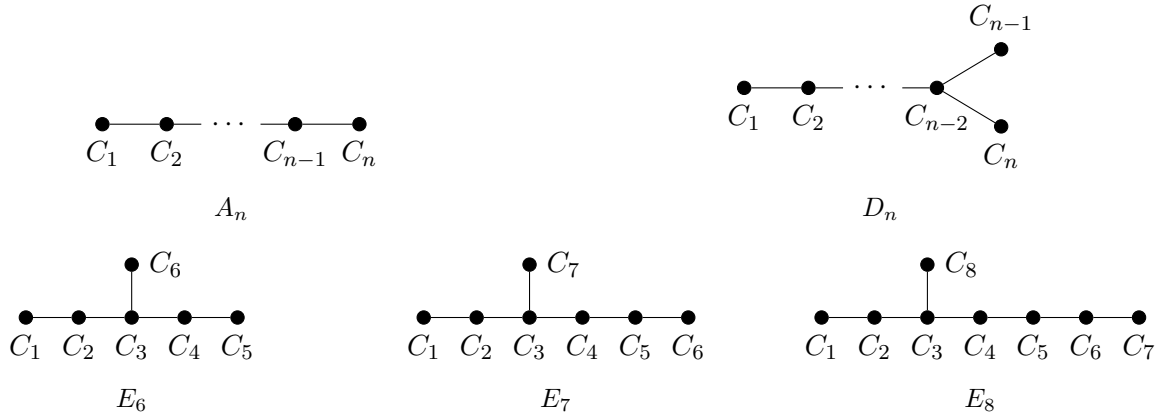


Figure 1: Dynkin diagrams / dual graphs of the exceptional curve configurations. Vertices are the (-2) -curves C_i ; an edge joins C_i, C_j whenever $C_i \cdot C_j = 1$.

Kodaira map preserve K3 structure. Writing $K_X = f^*K_Y + \sum_i a_i C_i$ and intersecting with C_j gives, since $K_X \cdot C_j = 0$ and $(f^*K_Y) \cdot C_j = 0$,

$$0 = \sum_i a_i (C_i \cdot C_j), \quad j = 1, \dots, r.$$

Negative definiteness of $(C_i \cdot C_j)$ forces $a_i = 0$ for all i , so f is crepant: $K_X = f^*K_Y$. Thus $\omega_Y \cong \mathcal{O}_Y$, and Y is a K3 surface with ADE (Du Val) singularities along $f(\bigcup_i C_i)$.

The image of the Kodaira map birationally stabilize, let us denote the image of Kodaira map $W_m := \Phi_{|mD|}(X)$.

Theorem 2.5 ([Uen75, Lemma 5.6]). There exists a positive integer m_0 such that, for any integer m satisfying $m \geq m_0$, $m \in \mathbb{N}(D, V)$, we have

$$\mathbb{C}(W_m) = \mathbb{C}(W_{m_0}).$$

That is, W_m is birationally equivalent to W_{m_0} for any $m \geq m_0$ such that $m \in \mathbb{N}(D, V)$. In particular, $\kappa(V, D) = \dim W_{m_0}$.

Proof. Let $d > 0$ be the greatest common divisor of the integers belonging to $\mathbb{N}(D, V)$. Since $\mathbb{N}(D, V)$ is a semigroup, there exists a positive integer ℓ_0 such that $\ell d \in \mathbb{N}(D, V)$ for every $\ell \geq \ell_0$. Set $k := \ell_0 d$, so that $nk \in \mathbb{N}(D, V)$ for every $n \geq 1$.

Fix $n \geq 1$ and let $\{\varphi_0, \dots, \varphi_N\}$ be a basis of $H^0(V, \mathcal{O}_V(nkD))$. The rational map

$$\Phi_{|nkD|}: V \dashrightarrow W_{nk} \subset \mathbb{P}^N, \quad x \mapsto [\varphi_0(x) : \dots : \varphi_N(x)],$$

identifies, on the affine chart $\{X_0 \neq 0\}$, the function field of W_{nk} with

$$\mathbb{C}(W_{nk}) = \mathbb{C}(X_1/X_0, \dots, X_N/X_0).$$

Pulling back by $\Phi_{|nkD|}$ gives an embedding

$$\Phi_{|nkD|}^*: \mathbb{C}(W_{nk}) \hookrightarrow \mathbb{C}(V), \quad X_i/X_0 \mapsto g_i := \varphi_i/\varphi_0,$$

so that $\mathbb{C}(W_{nk})$ is identified with the subfield $\mathbb{C}(g_1, \dots, g_N) \subset \mathbb{C}(V)$.

Now fix a nonzero element $\psi \in H^0(V, \mathcal{O}_V(kD))$, which exists since $k \in \mathbb{N}(D, V)$. The products $\psi\varphi_0, \dots, \psi\varphi_N$ lie in $H^0(V, \mathcal{O}_V((n+1)kD))$, and

$$\frac{\psi\varphi_i}{\psi\varphi_0} = \frac{\varphi_i}{\varphi_0} = g_i$$

for each i . Hence each g_i , viewed as an element of $\mathbb{C}(V)$, also lies in the (image of the) function field $\mathbb{C}(W_{(n+1)k})$. This yields an increasing chain of subfields of $\mathbb{C}(V)$,

$$\mathbb{C}(W_k) \subset \mathbb{C}(W_{2k}) \subset \mathbb{C}(W_{3k}) \subset \dots \subset \mathbb{C}(V).$$

By the Siegel–Remmert–Thimm theorem, the field $\mathbb{C}(V)$ of meromorphic functions on the compact complex manifold V has finite transcendence degree $d = \text{tr.deg}_{\mathbb{C}} \mathbb{C}(V) < +\infty$ over $\mathbb{C}$ and is finitely generated.

Set

$$L := \bigcup_{s \geq 1} \mathbb{C}(W_{sk}) \subseteq \mathbb{C}(V).$$

Then L is finitely generated over $\mathbb{C}$, say $L = \mathbb{C}(h_1, \dots, h_m)$ with $h_1, \dots, h_m \in L$. Since the chain $\{\mathbb{C}(W_{sk})\}_{s \geq 1}$ is increasing, there exists n_0 such that $h_1, \dots, h_m \in \mathbb{C}(W_{n_0k})$, whence $L \subseteq \mathbb{C}(W_{n_0k})$. Combined with the trivial inclusion $\mathbb{C}(W_{n_0k}) \subseteq L$, this gives

$$L = \mathbb{C}(W_{n_0k}).$$

Since the chain is increasing, for every $s \geq n_0$ we have

$$\mathbb{C}(W_{n_0k}) \subseteq \mathbb{C}(W_{sk}) \subseteq L = \mathbb{C}(W_{n_0k}),$$

so that $\mathbb{C}(W_{sk}) = \mathbb{C}(W_{n_0k})$ for all $s \geq n_0$.

Set $m_0 := (n_0 + 1)k$, and let $m \in \mathbb{N}(D, V)$ with $m \geq m_0$. Write

$$m = n_0k + m',$$

so $m' \geq k$. Since $d \mid k$ and $d \mid m$, we have $d \mid m'$; as $m'/d \geq k/d = \ell_0$, it follows that $m' \in \mathbb{N}(D, V)$.

Fix a nonzero element $\psi' \in H^0(V, \mathcal{O}_V(m'D))$, and let $\varphi_0, \dots, \varphi_N$ be a basis of $H^0(V, \mathcal{O}_V(n_0kD))$ as in Step 1, so that $g_i = \varphi_i/\varphi_0$. The products $\psi'\varphi_0, \dots, \psi'\varphi_N$ lie in $H^0(V, \mathcal{O}_V((n_0k + m')D)) = H^0(V, \mathcal{O}_V(mD))$, and on the chart $\{Y_0 \neq 0\}$ of W_m we have

$$\Phi_{|mD|}^* \left(\frac{Y_i}{Y_0} \right) = \frac{\psi'\varphi_i}{\psi'\varphi_0} = \frac{\varphi_i}{\varphi_0} = g_i.$$

Hence $g_i \in \mathbb{C}(W_m)$ for every i , and therefore

$$\mathbb{C}(W_{n_0k}) \subseteq \mathbb{C}(W_m). \quad (*)$$

For the reverse inclusion, since $m \in \mathbb{N}(D, V)$ the same multiplication argument as in Step 1 (multiplying a basis of $H^0(V, \mathcal{O}_V(mD))$ by a nonzero section of $H^0(V, \mathcal{O}_V((mk-m)D))$, noting $d \mid mk-m$ and $mk-m$ large enough to lie in $\mathbb{N}(D, V)$) gives

$$\mathbb{C}(W_m) \subseteq \mathbb{C}(W_{mk}).$$

Since $m \geq m_0 = (n_0 + 1)k > n_0$, the integer mk is of the form sk with $s = m \geq n_0$, so Step 2 gives

$$\mathbb{C}(W_{mk}) = \mathbb{C}(W_{n_0k}). \quad (**)$$

Combining $(*)$ and $(**)$,

$$\mathbb{C}(W_{n_0k}) \subseteq \mathbb{C}(W_m) \subseteq \mathbb{C}(W_{mk}) = \mathbb{C}(W_{n_0k}),$$

which forces

$$\mathbb{C}(W_m) = \mathbb{C}(W_{n_0k})$$

for every $m \in \mathbb{N}(D, V)$ with $m \geq m_0$. This completes the proof. $\square$

3 The Iitaka Fibration

We next introduce the Iitaka fibration, which is the limiting Kodaira map.

3.1 Existence of Iitaka fibrations

Theorem 3.1 (Iitaka fibration theorem, [Uen75]). Let D be a Cartier divisor on a variety V with $\kappa(D, V) > 0$. Then there exist a complex manifold V^* , a projective manifold W^* , and a surjective morphism $f: V^* \rightarrow W^*$ with the following properties.

- (1) There exists a modification $\pi: V^* \rightarrow V$.
- (2) $\dim W^* = \kappa(D, V)$.
- (3) For w in a dense subset U of W^* , in the complex topology, the fibre $V_w^* = f^{-1}(w)$ is irreducible and non-singular.
- (4) $\kappa(\pi^*D_w, V_w^*) = 0$ for every $w \in U$, where π^*D_w denotes the restriction of π^*D to V_w^* .

- (5) The fibre space $f: V^* \rightarrow W^*$ is bimeromorphically equivalent to the fibre space associated with the meromorphic map $\Phi_{mD}: V \dashrightarrow W_m$, for every integer $m \geq m_0$ with $m \in \mathbb{N}(D, V)$.

We may assume that V is normal, and, since $\kappa(D, V) > 0$, that D is effective.

Proof. Step 1: Resolving the Kodaira map. By Lemma 2.5 there exists $m_0 \in \mathbb{N}$ such that, for every $m \geq m_0$ with $m \in \mathbb{N}(D, V)$ and $W_m := \Phi_{mD}(V)$, one has $\dim W_m = \kappa(D, V)$ and $\mathbb{C}(W_m)$ is algebraically closed in $\mathbb{C}(V)$. Fix such an m and write $W := W_m$.

Let V^* be a non-singular model of the graph G of $\Phi_{mD}: V \dashrightarrow W$, and let $b: V^* \rightarrow V$ be the composite of the projection $G \rightarrow V$ with the desingularization $V^* \rightarrow G$; let $f: V^* \rightarrow W$ be the induced morphism, so that

$$\begin{array}{ccc} V^* & & \\ \downarrow b & \searrow f & \\ V & \dashrightarrow_{\Phi_{mD}} & W. \end{array}$$

Since b is a modification and V^*, V are normal, the pull back gives an isomorphism

$$b^*: H^0(V, \mathcal{O}_V(\ell D)) \xrightarrow{\sim} H^0(V^*, \mathcal{O}_{V^*}(\ell D^*))$$

for every $\ell > 0$, where $D^* = b^*D$; consequently f coincides with Φ_{mD^*} . We may therefore replace V^* by V from the outset, we may assume that $\Phi_{mD} = f$ is a morphism and V is non-singular.

Write $L := \mathcal{O}_V(mD)$. As will become clear, $f: V \rightarrow W$ is the Iitaka fibration.

Step 2: The relative Kodaira map over W . Set $N + 1 = \dim_{\mathbb{C}} H^0(V, L)$, so that $W \subset \mathbb{P}^N$. A hyperplane $\lambda_0 X_0 + \dots + \lambda_N X_N = 0$ cuts out a divisor H_λ on W , corresponding to a divisor $E_\lambda \in |mD|$. Let F be the fixed part of $|mD|$, put $E_\lambda^* = f^*H_\lambda$, so that $E_\lambda = E_\lambda^* + F$, and set $W_\lambda := W - H_\lambda$.

Since f is proper, $f_*(L^{\otimes n})$ is coherent on W for every $n > 0$; since W is projective, by GAGA, it will produce an algebraic coherent sheaf $\mathcal{F}_n$ on W^s (the associated scheme, we may omit it without causing any confusion) with $\mathcal{F}_n^{\text{an}} \cong f_*(L^{\otimes n})$.

Note that W_λ is affine, as it is a closed subscheme of $\mathbb{P}^N \setminus H_\lambda$. By Cartan A theorem $H^0(W_\lambda, \mathcal{F}_n)$ is generated, as an $H^0(W_\lambda, \mathcal{O}_{W_\lambda})$ -module, by finitely many sections $\psi_0, \dots, \psi_M$, as elements of

$$H^0(W_\lambda, f_*(L^{\otimes n})) = H^0(f^{-1}(W_\lambda), L^{\otimes n}).$$

Since

$$H^0(W_\lambda, \mathcal{F}_n) = H^0(W - H_\lambda, \mathcal{F}_n) \subset \bigcup_{e \geq 1} H^0(W, \mathcal{F}_n(eH_\lambda)),$$

and $\mathcal{F}_n(eH_\lambda) \cong \mathcal{F}_n \otimes \mathcal{O}_W([eH_\lambda])$ is coherent, we may choose e large enough that $\psi_i \in H^0(W, \mathcal{F}_n(enH_\lambda))$ for all i . By GAGA again,

$$H^0(W, \mathcal{F}_n(enH_\lambda)) \cong H^0(W, f_*(L^{\otimes n})(enH_\lambda)) = H^0(V, L^{\otimes n}(enE_\lambda^*)) \subset H^0(V, L^{\otimes n}(enE_\lambda)),$$

so each ψ_i may be regarded as a section in $H^0(V, L^{\otimes n}(enE_\lambda))$. Fixing $\eta \in H^0(V, \mathcal{O}_V(enH_\lambda))$ with $\text{div}(\eta) = enH_\lambda$, we obtain

$$\eta\psi_0, \dots, \eta\psi_M \in H^0(V, L^{\otimes(e+1)n}).$$

Let $\{\varphi_0, \dots, \varphi_N\}$ be a basis of $H^0(V, L)$, and let $\psi_{M+1}, \dots, \psi_{M'}$ be all monomials of degree n in the φ_j ; then $\eta\psi_i \in H^0(V, L^{\otimes(e+1)n})$ for $i = M+1, \dots, M'$ as well. These sections define a meromorphic map

$$h^{(n)}: V \dashrightarrow \mathbb{P}^{M'},$$

whose image we denote V_n . Since $\eta\psi_i \in H^0(V, L^{\otimes(e+1)n})$ for all $i = 0, \dots, M'$, and since $\{\varphi_j\}$ is a basis of $H^0(V, L)$, we get inclusions of function fields

$$\mathbb{C}(W) \subset \mathbb{C}(V_n) \subset \mathbb{C}(W_{(e+1)nm}),$$

where $W_{(e+1)nm}$ is the image of $\Phi_{(e+1)nmD}$. Since W already realizes the Kodaira dimension, $\mathbb{C}(W_{(e+1)nm}) = \mathbb{C}(W)$, so all three fields coincide. Hence there is a commutative diagram

$$\begin{array}{ccc} V & \xrightarrow{\quad h^{(n)} \quad} & V_n \\ & \searrow f \quad \swarrow g_n & \\ & W & \end{array}$$

with g_n birational.

Step 3: Kodaira dimension zero on a very general fibre. Since $f_*(L^{\otimes n})$ is coherent, there is a proper algebraic subset $S_n \subset W$ such that $f_*(L^{\otimes n})|_{W-S_n}$ is locally free, L is flat over $W - S_n$, and base change holds:

$$f_*(L^{\otimes n})(w) \xrightarrow{\sim} H^0(V_w, L_{V_w}^{\otimes n}), \quad w \in W - S_n.$$

For $w \in W_\lambda - S_n$, this identifies the restriction $\psi_{i,w}$ of ψ_i with an element of $H^0(V_w, L_w^{\otimes n})$.

By Corollary 1.8 (generic smoothness) there is an algebraic subset $T \subset W$ such that $W - T$ is non-singular and every fibre V_w , $w \in W - T$, is non-singular. Since W_λ is affine, $\mathcal{F}_n$ is globally generated there, so $H^0(V_w, L_w^{\otimes n})$ is spanned by $\psi_{i,w}$, $i = 0, \dots, M$, for every $w \in W_\lambda - S_n$.

Consequently, for $w \in W_\lambda - (S_n \cup T)$ the restriction $h_w^{(n)}: V_w \dashrightarrow \mathbb{P}^{M'}$ of $h^{(n)}$ is bimeromorphically equivalent to Φ_{mnD_w} , with image $h_w^{(n)}(V_w) = V_{n,w} = g_n^{-1}(w)$. (This holds because the $\psi_{i,w}$ span $H^0(V_w, L_w^{\otimes n})$ rather than merely lying in it, so $h_w^{(n)}$ agrees, up to a linear projection of the ambient projective space, with the map defined by a basis of $H^0(V_w, L_w^{\otimes n})$ — that is, with Φ_{mnD_w} .)

Since g_n is birational, there are nowhere dense algebraic subsets $B_n \subset V_n$ and $A_n \subset W$ such that g_n restricts to an isomorphism $V_n - B_n \xrightarrow{\sim} W - A_n$. Hence, for every $w \in W_\lambda - (S_n \cup T \cup A_n)$, the fibre $V_{n,w} = g_n^{-1}(w)$ is a single point, so that $h_w^{(n)}(V_w)$ is a point and

$$\dim_{\mathbb{C}} H^0(V_w, \mathcal{O}_{V_w}(mnD_w)) = 1.$$

Set $Y := H_\lambda \cup T \cup \bigcup_{n \geq 1} (S_n \cup A_n)$, a countable union of nowhere dense algebraic subsets of W , hence a set of the first category. Since W is a Baire space, $U := W - Y$ is dense in W . (Only density of U is used below; property (3) of the theorem does not require U to be open.) For every $w \in U$ and every $n \geq 1$,

$$\dim_{\mathbb{C}} H^0(V_w, \mathcal{O}_{V_w}(nmD_w)) = 1.$$

Since D is effective, $H^0(V_w, \mathcal{O}_{V_w}(nD_w)) \subset H^0(V_w, \mathcal{O}_{V_w}(nmD_w))$ for every n , so $\dim_{\mathbb{C}} H^0(V_w, \mathcal{O}_{V_w}(nD_w)) \leq 1$, and therefore

$$\kappa(D_w, V_w) = 0 \quad \text{for every } w \in U.$$

Moreover, since W is non-singular at each $w \in U$, Corollary 5.8 shows that V_w is irreducible.

Step 4: Desingularizing the base. Let $\delta: W^* \rightarrow W$ be a desingularization of W , obtained as a finite succession of monoidal transformations (Theorem 2.12). Then $g := \delta^{-1} \circ f: V \dashrightarrow W^*$ is a generically surjective meromorphic map, holomorphic on $V' := V - f^{-1}(T)$, since $W - T$ is non-singular.

Let G be the graph of g and let $\pi^*: V^* \rightarrow G$ be a desingularization of G ; set $\pi := p_V \circ \pi^*: V^* \rightarrow V$. Then π is a modification inducing an isomorphism $\pi^{-1}(V') \xrightarrow{\sim} V'$, and, since $W - T$ and $W^* - \delta^{-1}(T)$ are isomorphic, the induced morphism $f^* := f \circ \pi: V^* \rightarrow W^*$ together with $U^* := \delta^{-1}(U)$ satisfies properties (1)–(4) of the theorem.

□

3.2 Relative Iitaka Fibrations

Theorem 3.2 (Relative Iitaka fibration, [Nak04]). Let $f: X \rightarrow Z$ be a projective morphism of normal varieties, and let D be a $\mathbb{Q}$ -Cartier divisor on X . Let F be a general fiber of f and assume

$$r := \kappa(D|_F) \geq 0.$$

Then, after replacing X by a resolution $h: W \rightarrow X$, there is a contraction $g: W \rightarrow T$ over Z such that, if V is a general fiber of $T \rightarrow Z$ and G is a general fiber of g , then

$$\dim V = r, \quad \kappa(h^*D|_G) = 0.$$

$$\begin{array}{ccc} W & \xrightarrow{h} & X \\ g \downarrow & & \downarrow f \\ T & \longrightarrow & Z. \end{array}$$

Proof. Choose a sufficiently ample divisor A on Z so that

$$H^0(X, \mathcal{O}_X(mD + f^*A)) \neq 0$$

for some divisible $m > 0$, and set $L = D + f^*A$. Fujita's Iitaka dimension identity gives

$$\kappa(L) = \kappa(D|_F) + \dim Z = r + \dim Z.$$

Take the ordinary Iitaka fibration of L on a resolution of X :

$$\varphi: W \longrightarrow S.$$

We show that this fibration factors over Z . Let $\psi = f \circ h: W \rightarrow Z$. For a sufficiently divisible multiple, the linear system defining φ has the form

$$|kmL| = |M| + F_{\text{fix}}, \quad M \sim \varphi^*H$$

for an ample divisor H on S . Since $mD + f^*A$ has a nonzero section, after pullback to W we may write

$$mh^*D + \psi^*A \sim B \geq 0.$$

Thus

$$mh^*L \sim B + (m-1)\psi^*A.$$

For a large multiple k , subtracting the fixed part of $|kmh^*L|$ gives

$$M \sim N + k(m-1)\psi^*A$$

with $N \geq 0$. If C is a curve contracted by φ and chosen away from $\text{Supp } N$, then

$$0 = M \cdot C = N \cdot C + k(m-1)\psi^*A \cdot C.$$

Both terms on the right are non-negative, so $\psi^*A \cdot C = 0$. Since A is ample, $\psi(C)$ is a point. Thus curves contracted by φ are contracted by ψ on a dense open set, and the rigidity lemma gives a morphism $S \rightarrow Z$. We write this relative model as T , and write $g: W \rightarrow T$ for the induced contraction.

Finally,

$$\dim T = \kappa(L) = r + \dim Z,$$

so a general fiber V of $T \rightarrow Z$ has dimension r . On a general fiber G of g , the ordinary Iitaka theorem gives $\kappa(h^*L|_G) = 0$. Since G lies inside a fiber of ψ , the divisor ψ^*A restricts trivially to G . Hence $h^*L|_G = h^*D|_G$, and $\kappa(h^*D|_G) = 0$. $\square$

3.3 Universal properties of Iitaka fibrations

4 Several applications of Iitaka fibrations

4.1 Good Minimal Models and Abundance

The first application explains why the Iitaka fibration is not just a map attached to a linear system, but also a mechanism for reducing the abundance problem. Once the base has dimension κ and the very general fiber has Kodaira dimension zero, the remaining issue is whether the numerical positivity on the fiber disappears. If it does, one can often appeal to the known numerical-dimension-zero case and then lift the result back to the total space.

Theorem 4.1 (Good minimal model versus abundance). For pseudo-effective klt pairs, the existence of good minimal models is equivalent to abundance:

$$\kappa_\sigma(K_X + \Delta) = \kappa(K_X + \Delta).$$

Proof. Assume first that (X, Δ) has a good minimal model (X', Δ') . Then $K_{X'} + \Delta'$ is semiample, so it defines a morphism

$$f: X' \rightarrow Z = \text{Proj } R(X', K_{X'} + \Delta')$$

and

$$K_{X'} + \Delta' \sim_{\mathbb{Q}} f^*A$$

for an ample $\mathbb{Q}$ -divisor A on Z . Since Kodaira dimension and numerical Kodaira dimension are birational invariants for this purpose, and since semiample divisors are abundant,

$$\kappa_\sigma(K_X + \Delta) = \kappa_\sigma(K_{X'} + \Delta') = \dim Z = \kappa(K_{X'} + \Delta') = \kappa(K_X + \Delta).$$

Conversely assume abundance. If $\kappa_\sigma(K_X + \Delta) = \dim X$, then $K_X + \Delta$ is big and BCHM gives a good minimal model. If $\kappa_\sigma(K_X + \Delta) = 0$, the numerical-dimension-zero case gives a good minimal model. Thus we may assume

$$0 < \kappa(K_X + \Delta) = \kappa_\sigma(K_X + \Delta) < \dim X.$$

Let

$$f: X \dashrightarrow Z$$

be a birational model of the Iitaka fibration of $K_X + \Delta$, and let F be a very general fiber. The Iitaka fibration gives

$$\dim Z = \kappa(K_X + \Delta).$$

By the easy addition formula for numerical Kodaira dimension and its converse in this setting,

$$\kappa_\sigma(K_X + \Delta) = \kappa_\sigma(K_F + \Delta|_F) + \dim Z.$$

Therefore

$$\kappa_\sigma(K_F + \Delta|_F) = 0.$$

The numerical-dimension-zero case gives a good minimal model for the very general fiber. Applying the theorem that a variety whose Iitaka-fibration general fiber has a good minimal model itself has a good minimal model, one obtains a good minimal model for (X, Δ) . $\square$

4.2 Asymptotic Definition of Kodaira Dimension

Historically, Iitaka dimension was formulated through the growth of the number of sections. The Iitaka fibration gives the geometric reason for this growth: sections grow like sections on the base, while the very general fibers contribute only bounded-dimensional spaces of sections.

Theorem 4.2 ([Uen75]). Let V be a normal projective variety and let D be a Cartier divisor with $\kappa(D, V) \geq 0$. Let d be the greatest common divisor of $\mathbb{N}(D, V)$. Then there are positive constants α, β and an integer m_0 such that

$$\alpha m^{\kappa(D, V)} \leq h^0(V, \mathcal{O}_V(mdD)) \leq \beta m^{\kappa(D, V)}$$

for all $m \geq m_0$.

Proof. It is enough to prove the result after replacing D by a sufficiently large effective multiple $E \sim ndD$. Indeed, writing $m = m_1n + m_2$ with $0 \leq m_2 < n$, the estimates for E imply the corresponding estimates for D after changing the constants. Thus assume D is effective and that $\mathbb{N}(D, V) = \mathbb{N}$.

Choose D so that the Kodaira map

$$f = \Phi_{|D|}: V \rightarrow W$$

is a morphism, $\dim W = \kappa(D, V)$, and $\mathbb{C}(W)$ is algebraically closed in $\mathbb{C}(V)$. Let H_λ be a hyperplane section of $W \subset \mathbb{P}^N$, and let $E_\lambda \in |D|$ be the corresponding divisor on V . Write

$$E_\lambda = E_\lambda^* + F, \quad E_\lambda^* = f^*H_\lambda,$$

where F is the fixed part of $|D|$.

For the lower bound,

$$h^0(V, mD) = h^0(V, mE_\lambda) \geq h^0(V, mE_\lambda^*) \geq h^0(W, mH_\lambda).$$

Since $\dim W = \kappa(D, V)$ and H_λ is ample on W , the last term is bounded below by $\alpha m^{\kappa(D, V)}$ for $m \gg 0$.

For the upper bound, decompose the fixed part as

$$F = L + F^*,$$

where L is the sum of the components dominating W and F^* is vertical over W . Let

$$B_m = L_m + F_m^* \in |mD|$$

be a general member, with L_m the sum of components dominating W and F_m^* vertical. By the fiber statement in the Iitaka fibration theorem, for very general $w \in W$,

$$h^0(V_w, \mathcal{O}_{V_w}(mD|_{V_w})) = 1.$$

Restricting the linearly equivalent divisors B_m and mE_λ to such fibers gives

$$L_m|_{V_w} = mL|_{V_w}.$$

Since all components of L_m and L dominate W , this equality on very general fibers implies $L_m = mL$. Thus the horizontal part mL is fixed in $|mD|$, and

$$h^0(V, mD) = h^0(V, mF^* + mE_\lambda^*).$$

Every component of F^* maps to a proper subvariety of W , so after replacing an effective Cartier divisor H on W by a large multiple we have

$$F^* \leq f^*H.$$

Therefore

$$h^0(V, mD) \leq h^0(V, mf^*(H + H_\lambda)) = h^0(W, m(H + H_\lambda)).$$

The divisor $H + H_\lambda$ is effective on the $\kappa(D, V)$ -dimensional variety W , so the elementary growth estimate for effective divisors gives

$$h^0(W, m(H + H_\lambda)) \leq \beta m^{\kappa(D, V)}$$

for $m \gg 0$. This proves the upper bound. □

Corollary 4.3. For a line bundle L on a normal projective variety,

$$\kappa(X, L) = \limsup_{m \rightarrow \infty} \frac{\log h^0(X, L^{\otimes m})}{\log m}.$$

4.3 Easy Addition Formula

Theorem 4.4 (Easy addition). Let $f: V \rightarrow W$ be a fiber space between nonsingular projective varieties, and let D be a Cartier divisor on V . Then there is a dense open subset $U \subseteq W$ such that for every $w \in U$,

$$\kappa(D, V) \leq \dim W + \kappa(D_w, V_w), \quad V_w = f^{-1}(w), \quad D_w = D|_{V_w}.$$

Proof. If $\kappa(D, V) = -\infty$, there is nothing to prove. Fix $m \in \mathbb{N}(D, V)$ and set

$$\mathcal{L}_m = f_* \mathcal{O}_V(mD).$$

By generic freeness and base change, there is a dense open subset $U^{(m)} \subseteq W$ such that $\mathcal{L}_m|_{U^{(m)}}$ is locally free, V_w is nonsingular, and

$$\mathcal{L}_m \otimes \mathbb{C}(w) \simeq H^0(V_w, \mathcal{O}_{V_w}(mD_w))$$

for every $w \in U^{(m)}$.

Let

$$g^{(m)}: \mathbb{P}(\mathcal{L}_m|_{U^{(m)}}) \rightarrow U^{(m)}$$

be the associated projective bundle. The natural evaluation map defines a relative Kodaira map

$$h^{(m)}: f^{-1}(U^{(m)}) \dashrightarrow \mathbb{P}(\mathcal{L}_m|_{U^{(m)}})$$

over $U^{(m)}$. By the base change isomorphism above, the restriction

$$h_w^{(m)}: V_w \dashrightarrow \mathbb{P}(\mathcal{L}_m \otimes \mathbb{C}(w))$$

is exactly the Kodaira map $\Phi_{|mD_w|}$ for $w \in U^{(m)}$.

Shrinking $U^{(m)}$, we may assume that the dimension of the image of this fiberwise Kodaira map is constant. Then

$$\dim h^{(m)}(V) = \dim W + \dim \Phi_{|mD_w|}(V_w)$$

for w in a dense open subset. This is simply the fiber-dimension formula applied to the image of the relative Kodaira map over the base.

On the other hand, the canonical isomorphism

$$H^0(W, \mathcal{L}_m) = H^0(V, \mathcal{O}_V(mD))$$

gives a rational map

$$\mathbb{P}(\mathcal{L}_m) \dashrightarrow \mathbb{P}H^0(V, \mathcal{O}_V(mD))^\vee$$

whose composition with $h^{(m)}$ is the absolute Kodaira map $\Phi_{|mD|}$. Hence the image of $\Phi_{|mD|}$ is dominated by $h^{(m)}(V)$, so

$$\dim \Phi_{|mD|}(V) \leq \dim h^{(m)}(V) = \dim W + \dim \Phi_{|mD_w|}(V_w).$$

Now choose m sufficiently large so that $\dim \Phi_{|mD|}(V) = \kappa(D, V)$. Since

$$\dim \Phi_{|mD_w|}(V_w) \leq \kappa(D_w, V_w),$$

we obtain

$$\kappa(D, V) \leq \dim W + \kappa(D_w, V_w)$$

for w in a dense open subset of W . □

4.4 Finite Generation of the Canonical Ring

The previous applications use the Iitaka fibration as a birational map. Finite generation is stronger: it says that the stable pluricanonical maps are controlled by one finitely generated graded algebra. The proof below is written in the form suggested by Fujino's analytic proof [Fuj22]: the Iitaka fibration is the device that reduces finite generation to the big case on the base.

Theorem 4.5 (Finite generation of the canonical ring). Let (X, Δ) be a projective klt pair with $K_X + \Delta$ $\mathbb{Q}$ -Cartier. Then the log canonical ring

$$R(X, K_X + \Delta) = \bigoplus_{m \geq 0} H^0(X, \mathcal{O}_X(\lfloor m(K_X + \Delta) \rfloor))$$

is finitely generated.

Proof. We explain the reduction in the pseudo-effective case; the big case is the finite generation theorem of BCHM [BCHM10]. After taking a log resolution and replacing by a suitable birational model, assume that X is smooth and $\text{Supp } \Delta$ has simple normal crossings.

Let

$$f: X \dashrightarrow Z$$

be the Iitaka fibration of $K_X + \Delta$. After resolving the indeterminacy, we may assume that

$$f: X \rightarrow Z$$

is a morphism with connected fibers and that Z is smooth. The defining property of the Iitaka fibration is

$$\dim Z = \kappa(K_X + \Delta), \quad \kappa((K_X + \Delta)|_F) = 0$$

for a very general fiber F .

The canonical bundle formula expresses the adjoint divisor upstairs in terms of data on the base:

$$K_X + \Delta \sim_{\mathbb{Q}} f^*(K_Z + B_Z + M_Z)$$

after replacing Z by a suitable birational model. Here B_Z is the discriminant part and M_Z is the moduli part. Consequently, after taking a sufficiently divisible Veronese subring, the canonical ring of (X, Δ) is identified with

$$\bigoplus_{m \geq 0} H^0(Z, \mathcal{O}_Z(mk(K_Z + B_Z + M_Z)))$$

for some $k > 0$.

This is the point where the Iitaka fibration is essential. Since Z is the base of the Iitaka fibration,

$$\kappa(K_Z + B_Z + M_Z) = \dim Z.$$

Thus $K_Z + B_Z + M_Z$ is big. In other words, the Iitaka fibration has converted the original non-big divisor $K_X + \Delta$ on X into a big divisor on its canonical base.

It remains to replace the generalized pair on Z by an ordinary klt pair. Since M_Z is nef and $K_Z + B_Z + M_Z$ is big, write

$$K_Z + B_Z + M_Z \sim_{\mathbb{Q}} A + G$$

with A ample and $G \geq 0$. For $0 < \varepsilon \ll 1$, the pair

$$(Z, \Delta_Z), \quad \Delta_Z = B_Z + \varepsilon G + H$$

is klt for a general small effective $\mathbb{Q}$ -divisor $H \sim_{\mathbb{Q}} M_Z + \varepsilon A$, and

$$K_Z + \Delta_Z \sim_{\mathbb{Q}} (1 + \varepsilon)(K_Z + B_Z + M_Z).$$

By the truncation principle for finite generation, it is enough to prove finite generation for

$$R(Z, K_Z + \Delta_Z).$$

But $K_Z + \Delta_Z$ is big and (Z, Δ_Z) is klt, so BCHM applies. Hence $R(Z, K_Z + \Delta_Z)$ is finitely generated, and therefore so is $R(X, K_X + \Delta)$. $\square$

5 Further Directions

Remark 5.1 (Abundance). The Iitaka fibration is the natural map expected to be induced by a semiample canonical divisor. Thus abundance can be read as the statement that the canonical divisor descends from the base of the Iitaka fibration; see Nakayama [Nak04] and the finite-generation framework of BCHM [BCHM10].

Remark 5.2. Lai [Lai11] shows that, for terminal projective varieties, good minimal models of the general fibers of the Iitaka fibration imply a good minimal model of the total space. This is a standard way to reduce an intermediate Kodaira dimension problem to the Kodaira-dimension-zero fiber.

Remark 5.3 (Nef cotangent and nef anti-canonical geometry). In positivity problems, the Iitaka fibration often becomes more rigid. For example, under nef cotangent and semiample hypothesis, one expects the fibration to have torus-like fibers and a base with positive canonical class. This is a useful direction for connecting Iitaka fibrations with structure theorems for varieties with special tangent or anti-canonical positivity.

Remark 5.4 (Iitaka conjecture). Iitaka's conjecture predicts $\kappa(X) \geq \kappa(F) + \kappa(Y)$ for an algebraic fiber space $f: X \rightarrow Y$. The relative Iitaka fibration is one of the basic tools for separating the Kodaira growth along the fiber from the growth on the base; the maximal Albanese dimension case is treated by Hacon–Popa–Schnell [HPS18].

Remark 5.5 (Effective Iitaka fibration conjecture and boundedness). The existence theorem is qualitative: it says that the maps $\Phi_{|mK_X|}$ eventually stabilize. Effective Iitaka fibration and boundedness problems ask for uniform control of such an m , or of the varieties appearing in families with fixed Iitaka data; see Birkar–Zhang [BZ16].

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